

Isaac Au

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EDUCATION

Carnegie Mellon University

Pittsburgh, PA

B.S. in Computer Science, **GPA:** 3.9, Dean's List High Honors (2023, 2024, 2025)

Aug 2023 – May 2027

Coursework: Computer Vision (Graduate), Natural Language Processing (Graduate), Distributed Systems, Machine Learning, Computer Systems, Database Design, Application Development, Computational Finance

Teaching Assistant: Reasoning With Data (Fall 2025)

Clubs and Activities: Hong Kong Student Association President, CIA Buggy Social Chair, Tartanhacks

SKILLS

Programming Languages: Python, C, C++, Go (Golang), SQL, JavaScript, TypeScript, HTML/CSS, Bash/Shell, Swift

Frameworks & Tools: Claude Code (custom skills, MCPs, parallel orchestration), Cursor, Docker, AWS, Azure, Git, MongoDB, PostgreSQL, FastAPI (REST APIs), React, Vue

ML/AI Libraries: PyTorch, Pandas, NumPy, CUDA, LlamaFactory, Hugging Face, GPT/Gemini/Qwen/Llama

Languages: Fluent in English, Cantonese, Mandarin Chinese

WORK & RESEARCH EXPERIENCE

Ovis Medical Solutions

Hong Kong

Co-founder, Technical Lead

Feb 2024 – Feb 2026

- Designed a three-stage AI pipeline that chains LLM outputs through progressive refinement to produce severity classifications with confidence scores from unstructured user conversations.
- Architected a hybrid inference routing layer using a classifier to direct private queries to a local model (Google MedGemma) and complex reasoning to OpenAI, cutting API costs by ~40% and improving output quality.
- Built a REST API supporting 50,000+ multimodal data points across 1,000+ assessments with JWT auth, Twilio OTP, and CI/CD via Docker on Cloudflare and MongoDB Atlas.
- Secured \$100,000 in SAFE funding by leading a 5-person team through the technical MVP lifecycle.

Applied AI Research (Synthetic Data, LLM Reasoning & Evaluation)

Pittsburgh, PA

Researcher

May 2025 – Sept 2025

- Engineered a scalable synthetic data generation pipeline for medical reasoning, utilizing LLM-as-a-judge framework, autonomously filtering 2M+ raw clinical entries to curate a 50k-row high-fidelity instruction dataset.
- Orchestrated Supervised Fine-Tuning (SFT) for open-weight models (Qwen4B, 7B) via LlamaFactory, optimizing LoRA hyperparameters to boost medical reasoning accuracy and algorithmic transparency by ~15% over baseline.
- Architected an automated evaluation harness to benchmark 7+ foundation models (GPT-4o, Gemini), quantifying performance regressions and bottlenecks in Chain-of-Thought (CoT) to enforce safety in high-stakes diagnostics.

Rice Robotics

Hong Kong

Software Development Intern (Embodied AI)

May 2024 – Jul 2024

- Architected an LLM-driven autonomy system for an embodied AI robot using Azure OpenAI, building a multimodal pipeline to produce structured motor commands and real-time emotional display via sentiment analysis. Delivered to 10 investors within a 6-week MVP deadline.
- Slashed full round-trip latency by 3.5x (from ~10s to ~3s) by multithreading the multimodal inference pipeline and parallelizing sentiment analysis and motor command generation against streamed output tokens.

PROJECTS

DropIn — Real-Time Sensor Fusion Motion Capture on Consumer Devices

Feb 2026-Present

- Engineered a real-time mocap system using iOS IMU data as a low-cost alternative to \$50k+ optical setups, building a kinematic solver with quaternions to reconstruct rider kinematics without external cameras.
- Architected a distributed streaming backend that synchronizes state with <50ms latency, employing a custom burst-detection protocol to deliver real-time feedback faster than human reaction time.
- Developed a 3D telemetry and visualization suite using 3.js (WebGL) that translates multi-sensor data into a condensed form rating, filtering ~20% of noise to give users immediate, actionable coaching on their technique.

Distributed Worker Pool over Custom Reliable Transport Protocol

Oct 2025 – Nov 2025

- Engineered a reliable transport protocol (LSP) over UDP, utilizing sliding window flow control to sustain 100% data integrity across networks with simulated packet loss rates of up to 20%.
- Architected a parallelized worker pool using a Round Robin scheduler to increase throughput by ~300%, implementing heartbeat failure detection to automatically recover workers within 6 seconds of network partition.